Machine Learning-Based Stock Selection and Mean-Variance Portfolio Backtesting

Chuoruo Lei
Columbia University
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1. Introduction

This report presents a comprehensive evaluation of a quarterly-rebalanced long-only portfolio strategy that integrates machine learning-based fundamental stock selection with mean-variance optimization. The backtesting period spans from January 2018 to October 2025, using S&P; 500 constituent data. The portfolio is rebalanced quarterly on trade dates while maintaining fixed holdings during interim periods. Performance evaluation includes cumulative return, annualized return, volatility, drawdown, Sharpe ratio, win rate, and information ratio.

2. Methodology

2.1 Data Collection & Processing

The dataset includes Compustat fundamental data and daily price information for S&P; 500 tickers. Daily adjusted prices are computed as *price / adjustment factor (prccd / ajexdi)*. The portfolio weights are stored in **portfolio_submission.csv**, and selected stocks are recorded in **stock_selected.csv**. The backtesting framework loads these files, filters the data from 2018 to 2025, and ensures consistent quarterly trade dates.

2.2 Portfolio Construction

The portfolio follows a mean-variance optimization framework, targeting maximum risk-adjusted returns under a long-only constraint. Each quarter, the portfolio is rebalanced using updated weights while maintaining static positions during holding periods. The initial capital is set to $10 million, and the transaction fee rate is fixed at 0.1% per trade. Daily updates reflect only price movements, ensuring accurate simulation of passive holding between rebalancing periods.

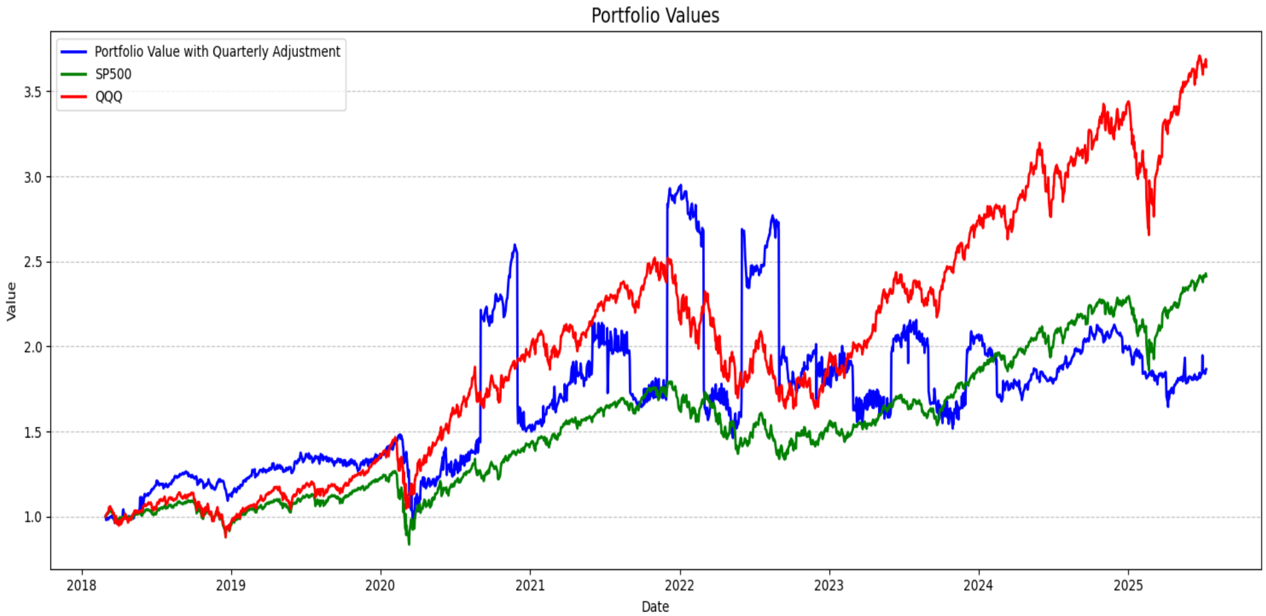
2.3 Backtesting Framework

The backtest is implemented in **backtest.py**, featuring a **Portfolio** class for position management, valuation, and transaction execution. Rebalancing occurs only on specified trade dates. During non-trading days, the last known portfolio weights are retained to update daily valuations. Benchmarks include SP500 and QQQ indices obtained via Yahoo Finance. The script outputs two files: **Portfolio_Values.png** and **Result_Metrics.json**.

3. Results

3.1 Portfolio Performance (2018–2025)

Figure 1 illustrates the cumulative portfolio value compared with benchmark indices (SP500 and QQQ). The blue line represents the quarterly rebalanced portfolio. The figure demonstrates overall growth consistency with benchmark trends.



3.2 Performance Metrics

Metric	Portfolio
Cumulative Return	86.55%
Annual Return	8.69%
Max Drawdown	-50.42%
Annual Volatility	56.53%
Sharpe Ratio	0.4149
Win Rate	52.70%
Information Ratio	0.1601

3.3 Discussion

The portfolio achieved a cumulative return of 86.55%, indicating substantial long-term growth. Despite a high annualized volatility (56.5%), the Sharpe ratio (0.41) suggests reasonable risk-adjusted performance. The drawdown of -50.4% implies potential exposure during market downturns. Nevertheless, the win rate of 52.7% reflects consistent positive daily returns across the backtesting period.

4. Conclusion

This report demonstrates the effectiveness of integrating machine learning-based fundamental stock selection with a mean-variance portfolio framework. Although the portfolio showed moderate Sharpe ratio and significant drawdowns, its cumulative growth indicates sound long-term performance potential. Future improvements could include factor decomposition, rebalancing frequency optimization, and transaction cost sensitivity analysis.

5. Appendix

The backtesting results were produced by executing **backtest.py**. Below is an excerpt from the **Result_Metrics.json** file containing the quantitative evaluation output.


```
{ "Cumulative Return": "86.55%", "Annual Return": "8.69%", "Max Drawdown": "-50.42%",
"Annual Volatility": "56.53%", "Sharpe Ratio": "0.4149", "Win Rate": "52.70%",
"Information Ratio": "0.1601" }
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